

**NO SUCH THING AS
EMPTY SPACE**

On the Planck scale,
space-time (see p.23)
may not be smooth and
empty but instead be
full of fluctuations.

**"In any field, find
the strangest thing
and then explore it."**

John Wheeler

**Space-time
bubbles**

Quantum gravity models predict that space-time is made up of tiny regions where dimensions froth in and out of existence, like bubbles in foam. This may be permitted by the uncertainty principle (see pp.42–43) over distances and intervals on the Planck scale.